Basic information

iAL curriculum **CUHK SBS team**	**HKDSE curriculum** **CUHK SBS team**	**Gaokao** **SUStech**	**Japanese High School Biology + AP Biology** **HG-Tokyo**
- Region / Curriculum system: Hong Kong international schools - Public examination system: Pearson Edexcel IAL Biology (Units/WBI11–WBI16; IAS + IA2). - School level / grade range: Typically S5–S6 (Grades 11–12 / post-16). - Subject name: Biology (International Advanced Level). - Compulsory or elective status: Elective, primarily for students pursuing science/medicine pathways. - Official curriculum documents: Pearson Edexcel IAL Biology Specification (2018 onwards). - Public examination papers: External written papers per unit + dedicated practical skills units (Unit 3 & 6); pre-released scientific articles in some papers. - Additional materials reviewed: CGP Head Start bridge book (foundational recap); revision notes and past paper examples on gene technology. The curriculum is modular, rigorous, and exam-focused, with strong emphasis on molecular mechanisms, biotechnology applications (especially in Unit 5), data interpretation, and practical skills. It excels in traditional molecular biology and gene technology but treats synthetic biology implicitly as advanced genetic engineering.	- Region / Curriculum system: Hong Kong local secondary schools - Public examination system: HKDSE Biology. - School level / grade range: Secondary 4–6 (S4–S6). - Subject name: Biology. - Compulsory or elective status: Compulsory part (all students) + Elective part (students choose any 2 out of 4 electives; Biotechnology is one of the four electives, ≈25 hours). - Official curriculum documents: Biology Curriculum and Assessment Guide (Secondary 4–6) 2007 (with updates in November 2015); Education Bureau subject overview documents. - Public examination papers: Paper 1 (Compulsory Part, 60%), Paper 2 (Elective Part, 20%), School-based Assessment (SBA, 20%). - Additional materials reviewed: Official curriculum guide sections on Molecular Genetics and Biotechnology elective; representative past-paper patterns; teaching resource summaries. The curriculum provides solid molecular biology foundations for all students and a dedicated Biotechnology elective covering recombinant DNA technology, PCR, cloning, GMOs, applications, and bioethics. It is strong in traditional biotechnology techniques and scientific inquiry but treats synthetic biology largely as an extension of genetic engineering rather than a distinct engineering discipline.	- Region / curriculum system: Chinese Mainland senior secondary biology and Gaokao Biology. - Public examination system: National College Entrance Examination (Gaokao), including biology in integrated science papers and provincial selective biology examinations under the new Gaokao model. - School level / grade range: Senior secondary school, mainly Grades 10-12, with Grade 12 students as public examination candidates. - Subject name: Biology / Gaokao Biology. - Compulsory or elective status: Under the new Gaokao model, Biology is usually a selective examination subject。. - Official curriculum documents: Senior High School Biology Curriculum Standards (2017 Edition, 2020 Revision); China Gaokao Evaluation System. - Public examination papers reviewed: Recent national and provincial Gaokao biology papers and textbooks. The curriculum is strongly oriented toward core biological concepts, experimental inquiry, data interpretation, and evidence-based reasoning. It provides strong foundations in molecular genetics and visible exposure to biotechnology tools such as PCR, electrophoresis, fermentation, and cell engineering. However, synthetic biology is rarely presented as an explicit curriculum or examination framework. It is usually embedded within genetic engineering, biotechnology, immunology, microbial applications, or innovation-oriented contexts rather than treated as a distinct design discipline.	- Region / Curriculum system: Japanese national high school curriculum (MEXT) + AP Biology for students in International / AP programs - Public examination system: Japanese university entrance examinations, including the Common Test + the AP Biology examination for students taking AP courses - School level / grade range: Typically S1–S3 / Grades 10–12 (approximately ages 15–18 - Subject name: Japanese high school Biology + AP Biology - Compulsory or elective status: selected according to the student's Japanese high school course of study, AP Biology is an elective/advanced course offered by participating schools - Official curriculum documents: MEXT High School Curriculum Guidelines and AP Biology Course and Exam Description - Public examination papers: Japanese university entrance examination papers, including Common Test Biology + AP Biology examinations - Additional materials reviewed: MEXT curriculum commentary and College Board AP Biology course, examination, science-practice, and laboratory requirements The combined curriculum is strong in molecular biology, gene expression, experimental reasoning, and data analysis. The Japanese curriculum provides systematic biological foundations and explicitly includes biotechnology and inquiry activities, while AP Biology adds college-level conceptual reasoning, scientific practices, and inquiry-based laboratory work. However, neither curriculum explicitly develops synthetic biology as a distinct engineering discipline.

Domain 1: Molecular Biology Foundations

iAL curriculum CUHK SBS team	HKDSE curriculum CUHK SBS team	Gaokao SUStech	Japanese High School Biology + AP Biology HG-Tokyo
Curriculum score: 3 (Advanced/Enrichment) **Examination score: 3**	**Curriculum score: 3 (Advanced/Enrichment)** **Examination score: 3**	**Curriculum score: 3 (Advanced / Enrichment)** **Examination score: 3**	**Curriculum score: 3 (Advanced / Enrichment)** **Examination score: 3**
Main content: Extensive coverage of biological molecules (proteins, carbohydrates, lipids, enzymes), DNA/RNA structure, replication, transcription, translation, gene expression regulation, mutations (substitution, insertion, deletion), genotype-phenotype relationships, and examples like cystic fibrosis. Students explain how DNA sequence changes affect RNA/protein structure/function and phenotype, and apply concepts to medical/biotech contexts. Gene regulation and epigenetics are implied in development topics. **Key evidence:** Units 1–2 (Molecules, Membranes, Proteins, DNA & Gene Expression); CGP bridge reinforces primary/secondary/tertiary protein structure, enzymes, and mutations. Exams require explanation, application, and data interpretation (e.g., sequence comparisons). **Comment:** Far exceeds essential/recommended levels. Strong foundation for synthetic biology designs (e.g., modifying gene expression or pathways).	**Main content:** Compulsory coverage of molecules of life (proteins, nucleic acids), cellular organisation, DNA structure, protein synthesis (transcription and translation), mutations, genotype–phenotype relationships, and basic recombinant DNA principles. Students explain how changes in DNA sequence can affect protein function and phenotype, and apply molecular concepts to biotechnology examples. **Key evidence:** Compulsory Part II – Genetics and Evolution (Molecular genetics); recommended practical work such as DNA extraction. Public examinations frequently test explanation, application, and data interpretation in this area. **Comment:** Exceeds essential and recommended levels. Provides a robust foundation that supports later synthetic biology applications (e.g., gene modification and expression control).	**Main content:** Extensive coverage of biological molecules, cells, DNA and RNA, DNA replication, transcription, translation, gene expression and regulation, mutations, heredity, genotype-phenotype relationships, and evolutionary mechanisms. Students explain how genetic information controls biological traits and how molecular-level changes affect proteins, functions, and phenotypes. **Key evidence:** Compulsory modules cover molecules and cells, genetics, evolution, and regulation. Gaokao papers frequently test molecular mechanisms through diagrams, experimental data, and unfamiliar biological contexts. **Comment:** Very strong foundation for synthetic biology. Students have the molecular knowledge needed to understand how DNA sequences, regulatory mechanisms, proteins, and phenotypes can be deliberately modified.	**Main content:** Extensive coverage of biological molecules, cells, DNA and RNA, DNA replication, transcription, translation, gene expression and its regulation, mutations, heredity, genotype–phenotype relationships, and developmental gene expression. The Japanese curriculum explicitly requires students to understand regulation of gene expression and connect transcriptional regulation with proteins. AP Biology provides additional college-level treatment of heredity and gene expression/regulation. **Key evidence:** MEXT biology includes gene expression, including gene expression regulation and the relationship between development and gene expression. AP Biology Units 5–6 cover Heredity and Gene Expression and Regulation, with gene regulation and molecular mechanisms assessed through application and analysis. **Comment:** Very strong foundation for synthetic biology. Students have the molecular knowledge necessary to understand how DNA sequences, regulatory mechanisms, proteins, and phenotypes can be deliberately modified.

Domain 2: Core Biotechnology Tools

iAL curriculum CUHK SBS team	HKDSE curriculum CUHK SBS team	Gaokao SUStech	Japanese High School Biology + AP Biology HG-Tokyo
Curriculum score: 3 **Examination score: 2–3**	**Curriculum score: 2–3 (3 for students taking the Biotechnology elective)** **Examination score: 2–3**	**Curriculum score: 2-3** **Examination score: 2-3**	**Curriculum score: 2–3** **Examination score: 2**
Main content: Recombinant DNA technology (restriction endonucleases, DNA ligase, plasmids as vectors, transformation), PCR (principles and uses), gel electrophoresis, DNA sequencing, genetic modification of microorganisms/plants/animals, cloning, and production of GM products (e.g., insulin, β-carotene in Golden Rice). Students explain steps, interpret results (e.g., gels), and compare techniques. **Key evidence:** Unit 5 (Gene Technology); core practicals and practical skills units. CGP bridge covers foundational tools indirectly via molecules/DNA. Exams test interpretation and simple workflows. **Comment:** Principles and applications are well-covered. Advanced workflows and troubleshooting are more implicit (stronger in practical assessments).	**Main content:** Compulsory: recombinant DNA technology, DNA fingerprinting, and the Human Genome Project. Elective (Biotechnology): detailed process of recombinant DNA technology (illustrated by insulin production), PCR and its applications, DNA fingerprinting, principles of producing genetically modified microorganisms/animals/plants, animal cloning (e.g., Dolly), and plant tissue culture. Students outline steps, recognise applications, and discuss benefits/hazards. **Key evidence:** Compulsory applied genetics + Elective VIII.a (Techniques in modern biotechnology). Recommended practicals include gel electrophoresis and PCR. Examinations test description of processes, interpretation of results, and applications. **Comment:** Principles and major tools are well covered, especially for elective students. Advanced workflow design and troubleshooting are more limited.	**Main content:** Genetic engineering, recombinant DNA technology, restriction enzymes, DNA ligase, plasmid vectors, transformation, PCR, electrophoresis, gene expression, fermentation engineering, cell engineering, cloning-related techniques, and biotechnology applications in medicine, agriculture, food production, and environmental contexts. Students interpret experimental results and apply biotechnology concepts to novel scenarios. **Key evidence:** Selective Compulsory 3, Biotechnology and Engineering, provides the clearest curriculum basis. Recent Gaokao papers include gene cloning, plasmid construction, PCR, electrophoresis, microbial engineering, fermentation, and plant or animal biotechnology. **Comment:** Principles and major applications are well covered. Advanced workflow design, troubleshooting, and construction of engineered biological systems are less consistently required.	**Main content:** Genetic technologies, recombinant DNA / genetic modification concepts, DNA analysis, electrophoresis, and biotechnology applications. MEXT explicitly identifies technologies involving genes and gives genetic recombination and DNA analysis using electrophoresis as examples of possible learning activities. AP Biology provides conceptual understanding of biotechnology-related processes and experimental methods, although advanced laboratory protocols are not the central purpose of the course. **Key evidence:** MEXT identifies techniques to work with genes, and gives genetic recombination technology and DNA analysis by electrophoresis as examples. AP Biology requires hands-on inquiry-based laboratories and scientific investigation skills. **Comment:** Good conceptual foundation, particularly when AP Biology and school-level laboratory work are combined. However, advanced workflows, troubleshooting, and actual construction of engineered biological systems are generally enrichment rather than core requirements.

Domain 3: Synthetic Biology Concepts

iAL curriculum
CUHK SBS team

Curriculum score: 1

Examination score: 1

Main content:
Gene technology and GMOs are presented as designing organisms for specific functions (e.g., drug production, nutrient enhancement). Concepts like biological parts (promoters, genes), vectors, and transgenic organisms appear, but there is no explicit distinction of synthetic biology from genetic engineering. No coverage of biological parts/devices/systems, chassis organisms, modularity, standardization (e.g., BioBricks), genetic circuits, input-output logic, or biosensors as synbio-specific.

Key evidence:
Unit 5 focuses on recombinant DNA, GM applications, and ethical implications. No mention of "synthetic biology," circuits, or standardization in the specification.

Comment:
Treated as advanced biotechnology. Weak on core synbio conceptual framework—curriculum remains biotechnology-oriented.

HKDSE curriculum
CUHK SBS team

Curriculum score: 1

Examination score: 1

Main content:
Gene technology and GMOs are presented as the design and construction of organisms for useful functions (e.g., pharmaceutical production, agricultural improvement). Biological components such as genes, vectors, and transgenic organisms appear, but there is no explicit distinction between synthetic biology and traditional genetic engineering. No coverage of biological parts/devices/systems, chassis organisms, modularity, standardisation, genetic circuits, input-output logic, or biosensors as synthetic-biology-specific concepts.

Key evidence:
Biotechnology elective focuses on recombinant DNA, GMOs, cloning, and applications. The term “synthetic biology” and related system-level concepts are absent from the curriculum guide.

Comment:
Treated as advanced biotechnology. Weak on the core conceptual framework that distinguishes synthetic biology.

Gaokao
SUStech

Curriculum score: 1

Examination score: 1

Main content:
Students encounter prerequisite concepts for synthetic biology, including genes, gene expression, genetic modification, biotechnology applications, and engineered organisms. However, synthetic biology itself is not established as a distinct conceptual framework. There is little explicit treatment of biological parts, devices, chassis organisms, modularity, standardization, BioBricks, genetic circuits, input-output logic, or biosensors as synthetic-biology-specific concepts.

Key evidence:
Curriculum and examinations focus mainly on molecular genetics, biotechnology, genetic engineering, fermentation, and biological applications. Synthetic biology-related ideas may appear indirectly, but the term and framework are not central.

Comment:
The curriculum is biotechnology-oriented rather than synthetic-biology-oriented. Students possess many prerequisites for synbio but are not systematically taught how biological components can be assembled into modular, designed systems.

Japanese High School Biology + AP Biology
HG-Tokyo

Curriculum score: 1

Examination score: 1

Main content:
Students encounter concepts that form the foundation of synthetic biology, including genes, gene regulation, genetic technologies, genetic modification, and biotechnology applications. However, synthetic biology itself is not established as a distinct conceptual framework. There is little or no explicit treatment of biological parts, devices, chassis organisms, modularity, standardization / BioBricks, genetic circuits, input-output logic, or biosensors as synthetic biology concepts.

Key evidence:
MEXT explicitly includes gene-related technologies and genetic recombination, but the curriculum materials do not establish a separate synthetic biology framework. AP Biology similarly covers gene expression, regulation, and biotechnology-related concepts but does not organize the course around synthetic biology systems or engineering concepts.

Comment:
The curriculum is biotechnology-oriented rather than synthetic-biology-oriented. Students possess many prerequisites for synbio but are not systematically taught how biological components can be assembled into modular, designed systems.

Domain 4: Design-Build-Test-Learn Thinking

iAL curriculum CUHK SBS team	HKDSE curriculum CUHK SBS team	Gaokao SUStech	Japanese High School Biology + AP Biology HG-Tokyo
Curriculum score: 1–2 **Examination score: 1–2** **Main content:** Iterative elements appear in practical design, hypothesis testing, data evaluation, and troubleshooting (e.g., enzyme experiments, gene tech workflows). Students suggest improvements based on results in practical papers, but there is no explicit DBTL cycle, open-ended biological system design, or project-based iteration. Gene tech involves "design" via gene selection/insertion but follows standard protocols. **Key evidence:** Practical Units 3 & 6; core practicals emphasize testing and controls. No iGEM-style or full engineering cycle in core content. **Comment:** Basic experimental logic and data-driven revision are present, but not framed as systematic engineering iteration.	**Curriculum score: 1–2** **Examination score: 1–2** **Main content:** Scientific inquiry is emphasised throughout (hypothesis formulation, identification of variables and controls, experimental design, data evaluation). SBA requires practical investigations. Students may suggest improvements based on results, but there is no explicit Design-Build-Test-Learn cycle, open-ended biological system design, or iterative engineering process. **Key evidence:** Scientific Inquiry strand across the curriculum; SBA practical-related tasks (20% of final grade). Written examinations include experimental design and evaluation questions, but these remain within traditional investigation formats. **Comment:** Basic experimental logic and data-driven revision are present, but the curriculum does not frame learning as systematic engineering iteration.	**Curriculum score: 1-2** **Examination score: 1-2** **Main content:** Scientific inquiry is strongly emphasised, including hypothesis testing, experimental design, variables and controls, data collection, result interpretation, and evaluation. Students often analyse procedures, identify errors, and suggest improvements. However, there is no explicit Design-Build-Test-Learn cycle, open-ended biological system design, or project-based engineering process. **Key evidence:** Curriculum standards emphasise scientific inquiry and scientific thinking. Gaokao papers frequently include experimental design and evaluation questions, but these remain within traditional investigation formats. **Comment:** Basic experimental logic and data-driven revision are present, but the curriculum does not frame learning as systematic engineering iteration.	**Curriculum score: 2** **Examination score: 2** **Main content:** Both systems include scientific inquiry, experimental design, hypothesis testing, data collection, interpretation, and evaluation. MEXT emphasizes observation, experimentation, inquiry, and development of students' ability to think, judge, and communicate scientifically. AP Biology explicitly requires scientific practices including Questions and Methods, data representation, statistical analysis, and argumentation. AP Biology also requires at least 25% of instructional time to involve hands-on, inquiry-based laboratory investigations. **Key evidence:** MEXT includes inquiry activities within biology topics, while AP Biology requires inquiry-based laboratories and six science practices. AP Biology's science practices include determining scientific questions and methods, analyzing data, and developing evidence-based arguments. **Comment:** Stronger than a purely lecture-based curriculum, but the full DBTL engineering cycle is not explicit. Students test hypotheses and revise explanations or experiments, but they are not normally required to design, build, test, and iteratively improve a synthetic biological system in the iGEM sense.

Domain 5: Data, Modelling, and Research Literacy

iAL curriculum
CUHK SBS team

Curriculum score: 2–3

Examination score: 2–3

Main content:
Excellent coverage of variables, controls (positive/negative), validity/reliability, graph/table interpretation, experimental design evaluation, repeatability, sample size, and evidence analysis. Bioinformatics, microarrays, and unfamiliar data (via pre-released articles) are included. Simple models (e.g., oxygen dissociation curves, Fick's Law) are used.

Key evidence:
Across all units, especially practical assessments and Unit 5. CGP bridge reinforces graphs, correlations, and planning experiments.

Comment:
Strong preparation for research; approaches recommended/advanced levels.

HKDSE curriculum
CUHK SBS team

Curriculum score: 2–3

Examination score: 2–3

Main content:
Strong emphasis on experimental design, identification of variables and controls, reliability and validity, graph and table interpretation, data analysis, and drawing evidence-based conclusions. SBA further develops these skills through practical investigations. Simple models appear in other topics (e.g., ecology, physiology).

Key evidence:
Compulsory scientific inquiry requirements; SBA assessment criteria; frequent data-interpretation questions in public examinations.

Comment:
Solid preparation for research literacy; approaches recommended/advanced levels, particularly through the combination of written papers and SBA.

Gaokao
SUStech

Curriculum score: 3

Examination score: 3

Main content:
Strong emphasis on graph and table interpretation, experimental design, variables and controls, model analysis, quantitative reasoning, evidence evaluation, prediction, explanation, and scientific argumentation. Students interpret unfamiliar biological information and connect data with mechanisms.

Key evidence:
The China Gaokao Evaluation System emphasises key abilities, disciplinary literacy, application, and innovation. Recent Gaokao biology questions frequently use experimental data, genetic diagrams, flow charts, models, and authentic scientific contexts.

Comment:
One of the strongest areas of the curriculum. Students are well prepared to interpret unfamiliar biological data and evaluate experimental evidence.

Japanese High School Biology + AP Biology
HG-Tokyo

Curriculum score: 3

Examination score: 3

Main content:
Experimental design, variables and controls, interpretation of tables and graphs, quantitative reasoning, evidence evaluation, model interpretation, statistical analysis, and scientific argumentation. AP Biology explicitly includes six science practices: concept explanation, visual representations, questions and methods, data representation, statistical tests / data analysis, and argumentation. MEXT also emphasizes scientific inquiry and the use of observations, experiments, and evidence to develop scientific reasoning.

Key evidence:
AP Biology allocates significant assessment weight to data analysis, visual representations, questions / methods, and argumentation; the free-response section includes experimental-result interpretation, graphing, scientific investigation, model analysis, and data analysis.

Comment:
One of the strongest areas of the combined curriculum. Students are well prepared to interpret unfamiliar biological data and evaluate experimental evidence, approaching the advanced / research-literacy level required for synthetic biology.

Domain 6: Bioethics, Biosafety, and Society

iAL curriculum
CUHK SBS team

Curriculum score: 2

Examination score: 2

Main content:
Benefits/risks of GMOs, ethical/social issues (e.g., genetic screening, gene therapy, GM crops/animals), stakeholder considerations, environmental release concerns, and regulation. Students discuss advantages/disadvantages and implications (e.g., Golden Rice risks). Biosafety is basic (containment implied in GM work).

Key evidence:
Unit 5 and genetic screening topics. Exams include evaluation from multiple viewpoints.

Comment:
Good balanced discussion at recommended level, but less structured advanced ethical analysis or dual-use concerns.

HKDSE curriculum
CUHK SBS team

Curriculum score: 2 (stronger for elective students)

Examination score: 2

Main content:
Explicit bioethics component in the Biotechnology elective covering ethical, legal, social, economic, and environmental issues related to genetically modified food, animal and plant cloning, the Human Genome Project, gene therapy, and stem-cell therapy. Students discuss benefits, hazards, and societal implications. Biosafety considerations appear mainly as risk/benefit discussion rather than structured risk assessment or containment protocols.

Key evidence:
Elective VIII.c (Bioethics); STSE connections throughout the curriculum. Examinations include evaluation and discussion-style questions on these issues.

Comment:
Good balanced treatment at the recommended level for elective students; less advanced structured ethical analysis or dual-use considerations.

Gaokao
SUStech

Curriculum score: 2

Examination score: 1-2

Main content:
Biotechnology is connected to food production, medicine, agriculture, environmental protection, biosafety, social responsibility, and ethical issues. Students may discuss benefits and risks of GMOs, cloning, gene editing, biological products, and biotechnology applications. However, formal synthetic-biology-specific risk assessment, containment design, dual-use concerns, and stakeholder analysis are limited.

Key evidence:
Biotechnology and Engineering includes links between biotechnology, society, safety, and ethics. Gaokao questions may include application and evaluation contexts, but extended ethical or biosafety analysis is not a major repeated examination focus.

Comment:
Good basic foundation for responsible biotechnology, but advanced biosafety and synthetic-biology-specific ethical analysis could be strengthened.

Japanese High School Biology + AP Biology
HG-Tokyo

Curriculum score: 2

Examination score: 1–2

Main content:
Biotechnology and genetic technologies can be considered in relation to their applications and social implications. Japanese education places biotechnology within real-world scientific contexts, while AP Biology requires students to apply biology to real-world questions and scenarios and become scientifically literate citizens. However, explicit synthetic-biology-specific treatment of containment, environmental release, dual-use concerns, stakeholder analysis, and formal risk assessment is limited.

Key evidence:
AP Biology requires opportunities to apply biology to real-world questions and scenarios. Japanese biotechnology education includes genetic technologies and their applications, while broader Japanese regulation provides a real-world context for biosafety and containment of genetically modified organisms.

Comment:
The combined system provides a reasonable foundation for responsible biotechnology, but advanced structured ethical analysis and synthetic-biology-specific biosafety are not central assessment targets.

Curriculum-Examination Alignment

iAL curriculum **CUHK SBS team**	**HKDSE curriculum** **CUHK SBS team**	**Gaokao** **SUStech**	**Japanese High School Biology + AP Biology** **HG-Tokyo**
Mostly Strong to Partial. Curriculum depth aligns well with exams, particularly in molecular foundations, tools, data, and ethics. Advanced design/iteration is better supported in practical components than written papers.	Mostly Strong to Partial. Curriculum depth is generally reflected in examinations, especially for molecular foundations, biotechnology tools (elective), data skills, and bioethics. Higher-order design and iterative thinking are better supported through SBA than through written papers. The elective nature of Biotechnology means alignment is stronger for students who choose it.	Mostly Strong to Partial. Curriculum and exams align strongly in molecular biology, genetics, experimental inquiry, data interpretation, and biological applications. Biotechnology tools are present in both curriculum and examinations, especially for students selecting Biology. However, explicit synthetic biology concepts, DBTL design thinking, genetic circuits, modularity, and project-based system construction are less consistently assessed.	Mostly Strong to Partial. Curriculum and exams align strongly in molecular biology and data / research literacy, but open-ended design, iterative engineering, and project-based synthetic biology are less consistently assessed.

Strengths of the Curriculum System

iAL curriculum **CUHK SBS team**	**HKDSE curriculum** **CUHK SBS team**	**Gaokao** **SUStech**	**Japanese High School Biology + AP Biology** **HG-Tokyo**
- Robust Domain 1 & 2 foundations and Domain 5 research skills. - Strong biotechnology applications and practical competence. - Good integration of ethics with real-world examples. - Prepares students effectively for university molecular biology/biotech.	- High level domain 1 & 2 foundations and Domain 5 research skills. - Explicit Biotechnology elective providing solid coverage of core tools, applications, and bioethics. - Strong scientific inquiry strand and 20% SBA weighting that support data literacy and practical competence. - Clear STSE emphasis linking biology to societal and ethical issues. - Prepares students effectively for university molecular biology/biotech.	- Strong Domain 1 molecular biology, genetics, gene expression, heredity, and regulation foundations. - Strong Domain 5 data interpretation, experimental design, evidence-based reasoning, and unfamiliar-context problem solving. - Visible biotechnology coverage, including PCR, electrophoresis, plasmids, genetic engineering, fermentation, and cell engineering. - Good preparation for university-level biology, medicine, agriculture, biotechnology, and life-science pathways.	- Strong Domain 1: Detailed molecular biology and gene expression knowledge. - Strong Domain 5: Strong data analysis, experimental design, and scientific reasoning. - Inquiry-based practical work: AP Biology emphasizes hands-on, inquiry-based labs. - Strong university preparation: AP Biology prepares students for advanced biology coursework.

Weaknesses and Gaps

iAL curriculum **CUHK SBS team**	**HKDSE curriculum** **CUHK SBS team**	**Gaokao** **SUStech**	**Japanese High School Biology + AP Biology** **HG-Tokyo**
- Domain 3: Explicit synthetic biology concepts (parts, devices, circuits, modularity, chassis, standardization) are absent or not distinguished. - Domain 4: Lacks explicit DBTL/project-based design thinking. - Limited system-level engineering, gene circuits, or iGEM-style open-ended work (enrichment-dependent). - Biosafety/risk assessment in Domain 6 is present but not advanced.	- Domain 3: Explicit synthetic biology concepts (biological parts/devices/systems, modularity, standardisation, chassis organisms, genetic circuits) are absent or not distinguished from genetic engineering. - Domain 4: Lacks explicit Design-Build-Test-Learn framing and open-ended biological system design. Biotechnology is elective only, so many students receive only basic recombinant DNA exposure. - Limited system-level engineering perspective and iGEM-style project work within the core curriculum. - Biosafety and risk assessment in Domain 6 remain at a relatively basic level.	- Domain 3: Explicit synthetic biology concepts such as biological parts, devices, systems, chassis organisms, modularity, standardization, and genetic circuits are limited. - Domain 4: Scientific inquiry is strong, but explicit DBTL cycles are not required. - Biotechnology is often assessed through interpretation rather than hands-on construction or open-ended design. - Limited iGEM-style project work within the formal examination curriculum. - Domain 6: Biosafety and ethics are present, but advanced risk assessment and dual-use discussion remain limited.	- Domain 3: Limited coverage of synthetic biology concepts like circuits, modularity, and chassis organisms. - Domain 4: Strong inquiry skills, but explicit DBTL cycles are not required. - Limited system-level engineering: Few opportunities to independently design biological systems. - Limited iGEM-style work: Open-ended synthetic biology projects are mainly extracurricular. - Domain 6: Basic biotechnology ethics are covered, but advanced biosafety and risk assessment could be strengthened.

Overall Synthetic Biology Readiness Judgment .

iAL curriculum **CUHK SBS team**	**HKDSE curriculum** **CUHK SBS team**	**Gaokao** **SUStech**	**Japanese High School Biology + AP Biology** **HG-Tokyo**
Moderate readiness. The iAL Biology curriculum delivers excellent molecular biology, biotechnology tools, data/research literacy, and responsible application awareness—stronger than many traditional programs and sufficient for university entry or basic biotech pathways. However, it is biotechnology-oriented rather than synthetic biology-oriented, with limited explicit engineering mindset, modularity, circuits, or iterative design. Students interested in iGEM or advanced synbio would benefit from extracurricular supplementation (e.g., school projects, online resources, or local initiatives).	Moderate readiness. The DSE Biology curriculum delivers excellent molecular biology foundations, biotechnology concepts, and data/research literacy making it a strong curriculum for biological pathways. However, it is biotechnology-oriented rather than synthetic biology-oriented similar to iAL biology, limiting students' ability to apply their knowledge to synthetic biological concepts. Students interested in iGEM or advanced synbio would benefit from extracurricular supplementation (e.g., school projects, online resources, or local initiatives). Furthermore, biotechnology is taught as part of an elective course, making students' literacy in technological tools unbalanced among the student body.	Moderate readiness. The Chinese Mainland senior secondary biology and Gaokao system provides strong foundations in molecular biology, genetics, biotechnology, experimental inquiry, and data-based reasoning. Students selecting Biology are generally well prepared to understand the biological mechanisms and analytical methods that support synthetic biology. However, the system remains more biology- and biotechnology-oriented than synthetic-biology-oriented, with limited focus on genetic circuits, modular design, chassis organisms, standardization, DBTL iteration, and open-ended engineering projects. Students interested in iGEM or advanced synthetic biology would benefit from extracurricular research, project-based enrichment, or university partnerships.	The Japanese high school + AP Biology curriculum provides strong foundations in molecular biology, biotechnology, experimental inquiry, data analysis, and scientific reasoning. However, it remains more biology- and biotechnology-oriented than synthetic-biology-oriented, with limited focus on genetic circuits, modular design, DBTL engineering, and iGEM-style projects. Therefore, it demonstrates Moderate Readiness, with extracurricular research or enrichment recommended for students pursuing synthetic biology.

Suggested Improvements (Realistic for High School)

iAL curriculum **CUHK SBS team**	**HKDSE curriculum** **CUHK SBS team**	**Gaokao** **SUStech**	**Japanese High School Biology + AP Biology** **HG-Tokyo**
- High priority: Add simple synbio case studies (biosensors, circuits) and DBTL-linked projects in Unit 5. - Medium priority: Strengthen modularity/standardization concepts and advanced ethical/risk analysis. - Develop optional enrichment modules or partnerships for iGEM-style work.	- High priority: Introduce simplified synthetic biology concepts (biological parts, devices, simple genetic circuits, modularity) and short DBTL-linked design tasks within the Biotechnology elective. - High priority: Strengthen data analysis and research literacy by making it a requirement in the curriculum, not an option. - Medium priority: Strengthen structured biosafety, risk assessment, and multi-stakeholder ethical analysis. - Medium priority: Add iGEM-style experiment design and analysis content. - Explore basics of biotechnology and synthetic biology as a compulsory subject. - Develop optional enrichment modules or school-based investigation pathways so that non-elective students also gain exposure to modern design-oriented ideas. - Encourage partnerships or project work that link curriculum content to iGEM-style activities.	- High priority: Add a short synthetic biology module covering biological parts, devices, chassis organisms, simple genetic circuits, biosensors, and modular design. - High priority: Introduce simple DBTL-linked design tasks within Biotechnology and Engineering. - Medium priority: Strengthen structured biosafety, risk assessment, and multi-stakeholder ethical analysis. - Medium priority: Add case studies on engineered microorganisms, environmental biosensors, metabolic engineering, and gene circuit applications. - Medium priority: Develop optional iGEM-style school projects or university-linked enrichment pathways.	- High priority: Add a short module on synthetic biology concepts and genetic circuits. - High priority: Introduce a simple DBTL-based design project. - Medium priority: Use biosensor and engineered-microorganism case studies. - Medium priority: Strengthen modularity, standardization, and biosafety concepts. - Medium priority: Offer optional iGEM / synthetic biology enrichment or university partnerships.